CHAPTER 15

Radiation Biology and Radiation Protection

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The biological effects of ionizing radiation were discovered soon after the production of intense radiation sources in the form of X-ray machines and radioactive elements about a century ago. Radiation was found to stop cell division and could therefore be used therapeutically to stop cancer growth. On the other hand, radiation applied locally was found to cause wounds, which were difficult to heal, and to induce cancer. Many serious accidents occurred as a result of the use of radiation before an adequate understanding of its biological effects led to formulation of rules for protection of workers. By 1922 approximately 100 radiologists (not patients) had died as a result of biological radiation damage.

The biological effects of *very large whole-body doses* are radiation sickness and early death, while *large organ doses* lead to local cell destruction and, possibly, organ death. The effects at *lower doses* are cell changes (decreased surviving fraction, decreased rate of division, chromosomal abberations, etc), which usually can be observed by microscope soon after irradiation. The induction of cancer may take years to observe and possible genetic changes may not be discovered until after several generations.

The creation of our world occurred in intense radiation fields. Consequently, we have inherited an Earth drenched in radiation from cosmic sources and the minerals in the ground (Ch. 12 and 13). Life evolved to survive in this normal "eco radiation". Essentially the current natural background levels are the lowest ever experienced by humankind. Though the intensity of these radiation sources is much smaller than produced by human techniques, no human can avoid these natural sources. In some cases the naturally occurring radioactive isotopes have been slightly enriched and taken out of their natural existence, e.g. use of uranium rich material in the building of houses resulting in a too radon rich environment inside the house. These increased levels have become important health issues. Closely related to this problem are the effects of man-made sources of similarly low levels, such as the storage of nuclear waste. Much research is presently devoted to the effects of low-level radiation.

Through extensive research in this field, our knowledge has grown enormously. Initially, the effects of radiation on local organs were studied; then the effects on various specific tissues and cell types were of concern; today, the focus is on the effects at the molecular level. This development is understandable as cancer induction and genetic changes are believed to have their roots in alterations in the DNA molecules in the cells. Table 15.1 lists a number of factors which affect the actual extent of the biological damage caused by radiation; most of these factors are discussed in this chapter. Many groups are actively engaged in research on the biological effects of ionizing radiation: *radiologists* use radiation for diagnosis and treatment of tumors; *health physicists* have the responsibility of controlling the use of irradiation equipment and protecting people from unnecessary exposure to radiation; in collaboration with *oncologists* (tumor researchers) and *geneticists*, *radiobiologists* conduct research to explain the effects of radiation on the cellular and molecular level, *radiation chemists* are interested in the interaction between radiation and the DNA molecule.

Table 15.1 Parameters in biological effects of radiation

Physical parameters
- Kind of radiation: α,β, or γ
- Internal or external source
- Size of total dose
- Dose rate
- Exposure time: instantaneous, temporary, recurrent, chronic

Microscopic effects
- molecular: DNA changes
- chromosome: abberations
- cellular (single or a few cells, *in vitro* or *in vivo*):
  - kind of cells (nervous system, bone marrow, liver, etc)
  - cell cycle (stage, rate of cell division, inactivation, etc)
  - temperature
  - oxygen content
  - cell poisons present (increasing radiation sensitivity)
  - protective agents present (anti-oxidants)
- tissue and organ: cancer growth or death

Macroscopic effects (inactivation, lethal dose (LD), cancer, etc.)
- Somatic: rapid (within a month), delayed (up to 20 years)
- Genetic: observed in offspring (one or several generations later)

15.1. THE BIOLOGICAL TARGET

The discussion of the effects of radiation on biological systems requires some familiarity with the target composition and common terminology.

The human body contains some 10^{14} cells. Figure 15.1.A shows schematically the cell and its nucleus, which contains thin thread-like *DNA-molecules* (*deoxyribonucleic acid*). The DNA carries the "genetic code" and is the most important molecule of an organism. It consists of two strands of sugar-phosphate chains, attached together by *base pairs* (forming so-called *nucleotides*), in a double helix form (Fig. 15.1.H–J). The human DNA contains 2.9×10^9 nucleotides; the DNA of simpler organisms has fewer nucleotides (down to about 5000). The nucleotides are combined in triplets (called *codons*), each one with the ability to produce (through some intermediate steps) a certain protein. The codons are ordered in long groups; the ordering is referred to as the *genetic code*. Because these long groups carry the information necessary to produce proteins for the different tissues, they are called *genes* ("makers"). In humans, the DNA is distributed over 23 pairs (altogether 46) *chromosomes*. In the chromosome, the DNA is wound around *histone* protein cores (Fig. 15.1.G) and highly twined ("condensed") to facilitate cell division; the DNA+histone unit (containing some 200 base-pairs) is referred to as the *chromatin*; these repeating units are known as *nucleosomes*. Figure 15.1.D shows some of the 23 chromosome pairs, and Figure 15.1.E a single one. The total amount of genetic